

Element B: Documentation and Analysis of Prior Design Attempts

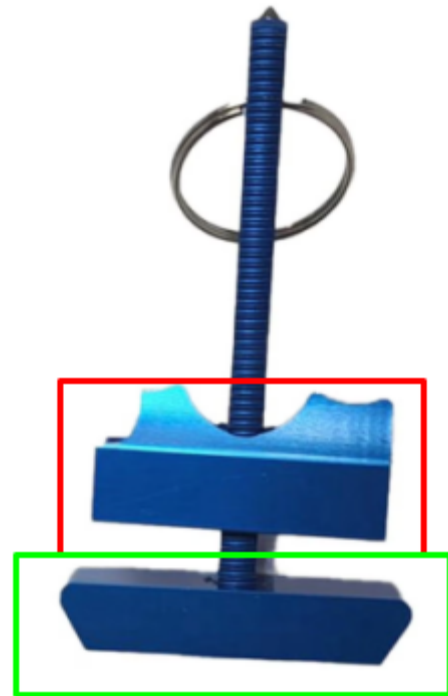
There are multiple prior attempts to solve the problem, as it affects people across the world. The first prior solution we found was the Stall Stopper.



The Stall Stopper (Source 1) fits on the top of the stall door and slides along it so the stall wall fits into the open section. This solution is very easily applicable and it is able to keep the door shut very effectively. However, there are a couple qualms. First, while being easy to remove is effective for the one using the toilet, it could also be removed through hard impact or someone on the outside of the stall. Secondly, the parents would also not “enjoy handling a device that's placed on unsanitary, bathroom stalls -- much less lugging it around in their pockets or purses.” (Business Insider, 2011). However, this issue will be prevalent for all solutions, regardless of how it is made.

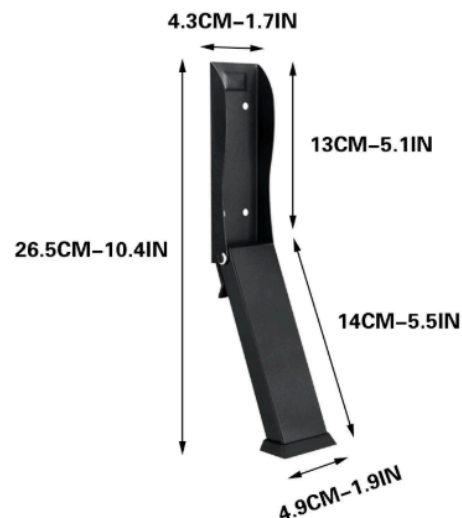
A second solution is the Stall Smart Safety Tool
(Source 2).

This product functions through the use of a screw. The user will widen the gap between the part boxed in red and the part boxed in green. The screw will go in the gap between the door and the stall wall, and then the red piece will be tightened against the door and the stall, locking the door in place. This will provide “safety with this heavy-duty steel lock, perfect for home, bedroom, apartment, office, or travel. Strong resistance against forced entry provides peace of mind and freedom from anxiety.” (Walmart). This product is built with strong materials, while also being compact, finding a balance of strength and transportability. One issue is that the tool doubles as a self-defense weapon, which causes accidental harm to the child or the parents. This product is sold at Walmart.



A third solution is the ZZkhGO No Drill Door
Stop Blocker (Source 3).

This product uses friction force to stop the door from opening. It attaches to the door through strong adhesives. With its compact



size of around 5.5" long and 1.7" wide, it is able to fit in the majority of men's pockets, and the vast majority of women's purses. The use of stainless steel allows the product to be strong and lightweight at the same time. However, this product was originally designed for regular doors, such as in a hotel room. This means that it may be too short to bridge the gap between the floor and the door. Additionally, the adhesive on the back of the lock may cause some minor damage to the door, including the paint tearing off of the door. This product is sold at Walmart.

A fourth solution is shown below (Solution 4):

This product works by hooking one end on the door and tightening the nut on the screw to push the black, rubber parts into the door. This clamps the door to the stall wall,

holding the door shut. The item has an approximate volume of 3.07 x 2.16 x 0.71in, which is clearly small enough to fit in a pocket or a purse.



Works Cited:

Source 1: MILLION-DOLLAR IDEA: A Clip That Prevents Restroom Doors From Opening, Business Insider,
<https://www.businessinsider.com/million-dollar-idea-a-clip-that-prevents-restroom-doors-from-opening-2011-1>, Jan. 6, 2011

Source 2: Stall Smart Safety Tool, Bathroom Lock Keychain, Portable Door for Hotel Rooms, Shaker (Blue), Walmart,
<https://www.walmart.com/ip/Stall-Smart-Safety-Tool-Bathroom-Lock-Keychain-Portable-Door-for-Hotel-Rooms-Shaker-Blue/15653069491>

Source 3: ZZkhGO No Drill Door Stop Blocker, Heavy Duty Stainless Steel Security Bar, Portable Travel Lock for Hotel Home Apartment Entry Protection, Walmart,
<https://www.walmart.com/ip/ZZkhGO-No-Drill-Door-Stop-Blocker-Heavy-Duty-Stainless-Steel-Security-Bar-Portable-Travel-Lock-for-Hotel-Home-Apartment-Entry-Protection/18837606482?wmlspartner=wlp&selectedSellerId=101633896>

Source 4: 2pcs Portable Door Lock Heavy Duty Home Security Door Lock Travel Hotel Room Lock, Walmart,
<https://www.walmart.com/ip/2pcs-Portable-Door-Lock-Heavy-Duty-Home-Security-Door-Lock-Travel-Hotel-Room-Lock/5252840274?wmlspartner=wlp&selectedSellerId=101561623>